

Software Requirements Specification

XPOS Management System

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Introduction

Purpose of the System

Issues cited related to a list of basic resource requirements that must be manually maintained by employees of a grocery store.

This includes handling their employee details, handling parts, keeping production records, and handling all good receipts and invoices. This process requires a lot of time and human effort and is also prone to human error.

The above matters take up a lot of time, and useful employees are wasted on trivial tasks. A product that helps in solving the above problems will prove to be beneficial to both the vendors and the consumers and will help in converting unproductive time into something more useful by eliminating unnecessary error complications and efforts.

Scope of the System

The aim of this project is to provide an efficient interface to manage their inventory based on each item sold. It will be an easy interface to manage the shop.

With this system, the cashier can easily get the invoice to the customer. At the end of each day, you can get a report on how many sales were made during the day, how much is the day's income and expenses and how much the remaining stock, what are the finished stocks, what are the returned products. And the generation of human resources is also considered here.

Databases and resources are updated with every transaction.

This system, combined with modern technology, brings satisfaction to both the trader and the customer.

Definitions, Acronyms and Abbreviations

DBMS – Database Management System

GUI – Graphical User Interface

OS – Operating System

SRS – Software Requirements Specification

SQL – Server Query Language

References and Acknowledgments

This specification references or complies with the following documents:

www.utdallas.edu

www.turtorialspoint.com

www.techtarget.com

www.microtoolsinc.com

Technologies to be used

- Java language
- Structured Query Language
- Jasper report
- Database Management System

Tools to be used

- NetBeans
- PhpMyAdmin
- Workbench
- Jasper soft studio

Overall Description

The overall description of our project can be creating and managing the database, developing a friendly user interface to manipulate the database, provide an authentication mechanism to safely accomplish tasks mentioned above.

Product Perspective

We are introducing a new system for grocery shops. When using the system, if an employee is hired, enter the role type of that employee and then create a user account, mark the data that can be accessed by that employee in the system and enter the username and password. Then each of those users enters the authentication page and after that, the system will authorize them. Users will basically query and edit the database through the system after getting permission according to their permission (role type).

User and Characteristics

Can add desired rolls as needed.

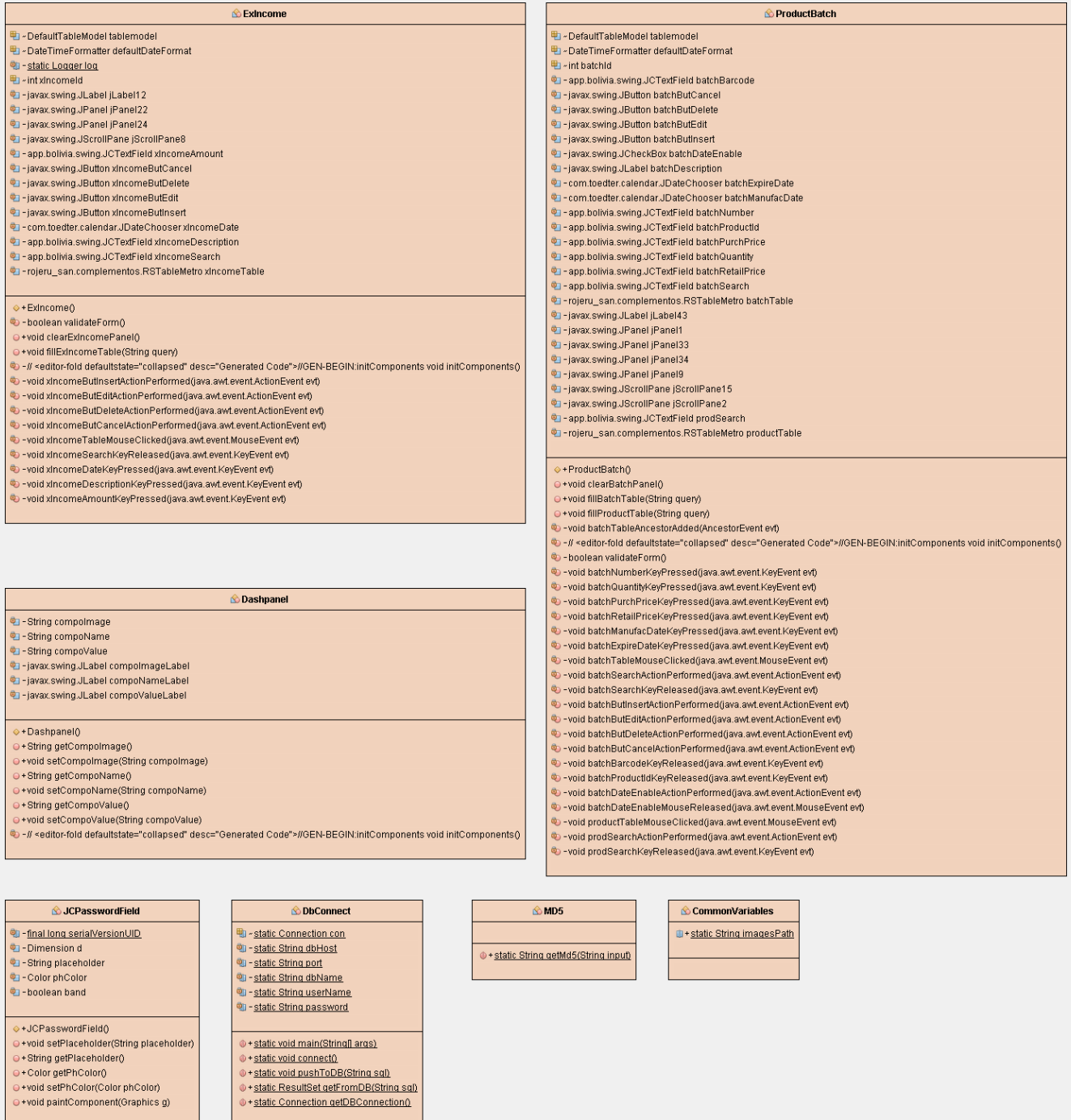
Ex: -

- **Cashier**
Operate the system to insert customers, suppliers and manage them.
- **Data Entry Operator**
Will Insert Employee details and also insert Items to the stock.

Design and Implementation Constraints

The application will be storing 1000 items data and employees' data. Roughly 200MB of storing capacity is needed. The software will be used on PCs. Application interfaces will be developed by java (SE) framework. Software will support JDK 8 environment and software personal database model will support MYSQL environment as DBMS. The system will run on only platform supporting JDK technology. Application will run on 256MB or higher of RAM.

Class Diagram



SystemUser
- static int userID - static String userName - static boolean sale - static boolean purchase - static boolean returns - static boolean customer - static boolean supplier - static boolean employee - static boolean userPosition - static boolean userProfile - static boolean products - static boolean exIncome - static boolean exCost - static boolean invoice - static boolean report
+ static int getUserid() + static void setUserid(int userID) + static String getUserName() + static void setName(String userName) + static boolean isSale() + static void setSale(boolean sale) + static boolean isPurchase() + static void setPurchase(boolean purchase) + static boolean isReturns() + static void setReturns(boolean returns) + static boolean isCustomer() + static void setCustomer(boolean customer) + static boolean isSupplier() + static void setSupplier(boolean supplier) + static boolean isEmployee() + static void setEmployee(boolean employee) + static boolean isUserPosition() + static void setUserPosition(boolean userPosition) + static boolean isUserProfile() + static void setUserProfile(boolean userProfile) + static boolean isProducts() + static void setProducts(boolean products) + static boolean isExIncome() + static void setExIncome(boolean exIncome) + static boolean isExCost() + static void setExCost(boolean exCost) + static boolean isInvoice() + static void setInvoice(boolean invoice) + static boolean isReport() + static void setReport(boolean report) + static void user(int userID, String userName)

Customer
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - static Logger log - int customerId - app.bolivia.swing.JTextField customerAddress - javax.swing.JButton customerButCancel - javax.swing.JButton customerButDelete - javax.swing.JButton customerButEdit - javax.swing.JButton customerButInsert - com.toedter.calendar.JDateChooser customerDOB - app.bolivia.swing.JTextField customerEmail - javax.swing.JRadioButton customerGenFemale - javax.swing.JRadioButton customerGenMale - com.toedter.calendar.JDateChooser customerJoinedDate - app.bolivia.swing.JTextField customerNIC - app.bolivia.swing.JTextField customerName - app.bolivia.swing.JTextField customerSearch - rojeru_san.complementos.RSTableMetro customerTable - app.bolivia.swing.JTextField customerTeleNumber - javax.swing.ButtonGroup gender - javax.swing.JLabel jLabel7 - javax.swing.JPanel jPanel6 - javax.swing.JPanel jPanel7 - javax.swing.JPanel jPanel8 - javax.swing.JScrollPane jScrollPane3
+ Customer() - boolean validateForm() - void clearCustomerPanel() - void fillCustomerTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void customerButInsertActionPerformed(java.awt.event.ActionEvent evt) - void customerButEditActionPerformed(java.awt.event.ActionEvent evt) - void customerButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void customerButCancelActionPerformed(java.awt.event.ActionEvent evt) - void customerTableMouseClicked(java.awt.event.MouseEvent evt) - void customerSearchKeyReleased(java.awt.event.KeyEvent evt) - void customerNameKeyReleased(java.awt.event.KeyEvent evt) - void customerAddressKeyPressed(java.awt.event.KeyEvent evt) - void customerTeleNumberKeyPressed(java.awt.event.KeyEvent evt) - void customerNICKeyPressed(java.awt.event.KeyEvent evt) - void customerNameKeyPressed(java.awt.event.KeyEvent evt) - void customerDOBKeyPressed(java.awt.event.KeyEvent evt) - void customerJoinedDateKeyPressed(java.awt.event.KeyEvent evt) - void customerEmailKeyPressed(java.awt.event.KeyEvent evt) - void customerGenMaleKeyPressed(java.awt.event.KeyEvent evt) - void customerGenFemaleKeyPressed(java.awt.event.KeyEvent evt) - void customerAddressKeyReleased(java.awt.event.KeyEvent evt)

Validate
+ static boolean isName(String in) + static boolean isNumber(String in) + static boolean isDoubleNumber(String in) + static boolean isNIC(String in) + static boolean isText(String in) + static boolean isEmail(String email) + static boolean isTelephone(String in) + static boolean isDate(String date)

Report
+ static JasperDesign design + static JasperReport saleReport + static JasperReport returnReport80mm + static JasperReport returnReport + static JasperReport RFQReport + static JasperReport POReport + static JasperReport purchaseReport
+ static void compile()

SystemLogger
+ static void initLogger()

Supply
- DefaultTableModel tablemodel - int supplierid - javax.swing.JLabel jLabel5 - javax.swing.JPanel jPanel11 - javax.swing.JPanel jPanel12 - javax.swing.JScrollPane jScrollPane1 - app.bolivia.swing.JTextField supAddress - javax.swing.JButton supButCancel - javax.swing.JButton supButDelete - javax.swing.JButton supButEdit - javax.swing.JButton supButInsert - app.bolivia.swing.JTextField supName - app.bolivia.swing.JTextField supRegisterNumber - app.bolivia.swing.JTextField supSearch - app.bolivia.swing.JTextField supTeleNumber - rojeru_san.complementos.RSTableMetro supplierTable
+ Supply() - boolean validateForm() - void clearSupPanel() - void fillSupTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void supButInsertActionPerformed(java.awt.event.ActionEvent evt) - void supButEditActionPerformed(java.awt.event.ActionEvent evt) - void supButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void supButCancelActionPerformed(java.awt.event.ActionEvent evt) - void supplierTableMouseClicked(java.awt.event.MouseEvent evt) - void supSearchKeyReleased(java.awt.event.KeyEvent evt) - void supNameKeyPressed(java.awt.event.KeyEvent evt) - void supAddressKeyPressed(java.awt.event.KeyEvent evt) - void supTeleNumberKeyPressed(java.awt.event.KeyEvent evt) - void supRegisterNumberKeyPressed(java.awt.event.KeyEvent evt) - void supNameKeyReleased(java.awt.event.KeyEvent evt)

POInvoice
- static org.apache.log4j.Logger log - rojeru_san.complementos.RSTableMetro POItemTable - app.bolivia.swing.JTextField POSearchByInvoice - app.bolivia.swing.JTextField POSearchBySupplier - rojeru_san.complementos.RSTableMetro POTable - javax.swing.JButton butInvoicePrint - javax.swing.JLabel jLabel6 - javax.swing.JPanel jPanel11 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane2 - javax.swing.JScrollPane jScrollPane3
+ POInvoice() + void fillPOInvoiceTable(String query) - void fillPOItemTable(int POid) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void POTableMouseClicked(java.awt.event.MouseEvent evt) - void POSearchByInvoiceActionPerformed(java.awt.event.ActionEvent evt) - void POSearchByInvoiceKeyReleased(java.awt.event.KeyEvent evt) - void POItemTableMouseClicked(java.awt.event.MouseEvent evt) - void POSearchBySupplierActionPerformed(java.awt.event.ActionEvent evt) - void POSearchBySupplierKeyReleased(java.awt.event.KeyEvent evt) - void butInvoicePrintActionPerformed(java.awt.event.ActionEvent evt)

User
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - int brandid - javax.swing.JLabel addressValue - javax.swing.JLabel dobValue - javax.swing.JLabel emailValue - javax.swing.JLabel genderValue - javax.swing.JLabel idValue - javax.swing.JLabel jLabel1 - javax.swing.JLabel jLabel10 - javax.swing.JLabel jLabel12 - javax.swing.JLabel jLabel14 - javax.swing.JLabel jLabel16 - javax.swing.JLabel jLabel18 - javax.swing.JLabel jLabel20 - javax.swing.JLabel jLabel3 - javax.swing.JLabel jLabel5 - javax.swing.JLabel jLabel6 - javax.swing.JLabel jLabel8 - javax.swing.JPanel jPanel1 - javax.swing.JPanel jPanel18 - javax.swing.JLabel jdValue - javax.swing.JLabel nameValue - javax.swing.JLabel nicValue - javax.swing.JLabel positionValue - javax.swing.JLabel telephoneValue - javax.swing.JButton userProButCancel - javax.swing.JButton userProButEdit - javax.swing.JPasswordField userProConfirmPassword - javax.swing.JPasswordField userProPassword - app.bolivia.swing.JTextField userProUsername
+ User() + void clearForm() - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void userProButEditActionPerformed(java.awt.event.ActionEvent evt) - void userProButCancelActionPerformed(java.awt.event.ActionEvent evt)

PurchaseOrder
- int supplierid - DefaultTableModel tablemodel - DateFormatter defaultDateFormat - DateFormatter defaultTimeFormat - Double totalWithNoDiscount - Double discount - Double finalTotal - int purchaseOrderid - javax.swing.ButtonGroup buttonGroup1 - javax.swing.JLabel jLabel13 - javax.swing.JLabel jLabel14 - javax.swing.JLabel jLabel40 - javax.swing.JLabel jLabel41 - javax.swing.JLabel jLabel42 - javax.swing.JPanel jPanel25 - javax.swing.JPanel jPanel26 - javax.swing.JPanel jPanel3 - javax.swing.JPanel jPanel5 - javax.swing.JScrollPane jScrollPane1 - javax.swing.JScrollPane jScrollPane9 - javax.swing.JButton purchaseOrderAddToTable - javax.swing.JLabel purchaseOrderBrand - javax.swing.JButton purchaseOrderCancel - javax.swing.JButton purchaseOrderComplete - javax.swing.JButton purchaseOrderDeleteItem - javax.swing.JLabel purchaseOrderDescription - javax.swing.JRadioButton purchaseOrderDisAmount - javax.swing.JRadioButton purchaseOrderDisPercent - app.bolivia.swing.JTextField purchaseOrderDiscount - javax.swing.JLabel purchaseOrderFinalDiscount - javax.swing.JLabel purchaseOrderItemId - app.bolivia.swing.JTextField purchaseOrderProdDescription - rojeru_san.complementos.RSTableMetro purchaseOrderProdTable - rojeru_san.complementos.RSTableMetro purchaseOrderProduct - app.bolivia.swing.JTextField purchaseOrderProductId - javax.swing.JLabel purchaseOrderProductTotal - app.bolivia.swing.JTextField purchaseOrderPurchasePrice - app.bolivia.swing.JTextField purchaseOrderQuantity - javax.swing.JLabel purchaseOrderSubTotal - javax.swing.JComboBox<String> purchaseOrderSupplierCombo - javax.swing.JLabel purchaseOrderTotal
+ PurchaseOrder() - void loadPurchaseOrderSupplierCombo() - void fillPurchaseOrderProdTable(String query) + callPurchaseOrderDetails() - void clearPurchaseOrderPanel() - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void purchaseOrderSupplierComboActionPerformed(java.awt.event.ActionEvent evt) - void purchaseOrderProductIdKeyReleased(java.awt.event.KeyEvent evt) - void purchaseOrderQuantityKeyReleased(java.awt.event.KeyEvent evt) - void purchaseOrderAddToTableActionPerformed(java.awt.event.ActionEvent evt) - void purchaseOrderProductMouseClicked(java.awt.event.MouseEvent evt) - void purchaseOrderCompleteActionPerformed(java.awt.event.ActionEvent evt) - void printInvoice(int invoiceid) - void purchaseOrderDeleteItemActionPerformed(java.awt.event.ActionEvent evt) - void purchaseOrderCancelActionPerformed(java.awt.event.ActionEvent evt) - void purchaseOrderDiscountKeyReleased(java.awt.event.KeyEvent evt) - void purchaseOrderDisAmountMouseClicked(java.awt.event.MouseEvent evt) - void purchaseOrderDisPercentMouseClicked(java.awt.event.MouseEvent evt) - void purchaseOrderProdTableMouseClicked(java.awt.event.MouseEvent evt) - void purchaseOrderProdDescriptionKeyReleased(java.awt.event.KeyEvent evt) - void purchaseOrderPurchasePriceKeyReleased(java.awt.event.KeyEvent evt)

QuateInvoice
- javax.swing.JButton butInvoicePrint - javax.swing.JButton butSetResponded - javax.swing.JLabel jLabel13 - javax.swing.JScrollPane jScrollPane2 - javax.swing.JScrollPane jScrollPane3 - rojeru_san.complementos.RSTableMetro quateItemTable - rojeru_san.complementos.RSTableMetro quateTable - app.bolivia.swing.JTextField saleSearchByCustomer - app.bolivia.swing.JTextField saleSearchByInvoice
+ QuateInvoice() + void fillQuateInvoiceTable(String query) - void fillQuateItemTable(int quateid) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void saleSearchByInvoiceActionPerformed(java.awt.event.ActionEvent evt) - void saleSearchByInvoiceKeyReleased(java.awt.event.KeyEvent evt) - void saleSearchByCustomerActionPerformed(java.awt.event.ActionEvent evt) - void saleSearchByCustomerKeyReleased(java.awt.event.KeyEvent evt) - void quateTableMouseClicked(java.awt.event.MouseEvent evt) - void quateItemTableMouseClicked(java.awt.event.MouseEvent evt) - void butSetRespondedActionPerformed(java.awt.event.ActionEvent evt) - void butInvoicePrintActionPerformed(java.awt.event.ActionEvent evt)

SupplierProduct
- DefaultTableModel tablemodel - app.bolivia.swing.JTextField itemIdCode - javax.swing.JLabel jLabel5 - javax.swing.JPanel jPanel1 - javax.swing.JPanel jPanel11 - javax.swing.JPanel jPanel12 - javax.swing.JPanel jPanel13 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane1 - javax.swing.JScrollPane jScrollPane2 - javax.swing.JScrollPane jScrollPane3 - app.bolivia.swing.JTextField productId - javax.swing.JLabel productName - javax.swing.JButton supProdCancel - javax.swing.JButton supProdDelete - javax.swing.JButton supProdEdit - javax.swing.JButton supProdInsert - app.bolivia.swing.JTextField supProdProductSearch - rojeru_san.complementos.RSTableMetro supProdProductTable - app.bolivia.swing.JTextField supProdSupplierSearch - rojeru_san.complementos.RSTableMetro supProdSupplierTable - app.bolivia.swing.JTextField supProductSearch - app.bolivia.swing.JTextField supProductid - javax.swing.JLabel supplierName - rojeru_san.complementos.RSTableMetro supplierProductTable
+ SupplierProduct() - void clearSupplierProductPanel() + void fillProductTable(String query) - void fillSupplierTable(String query) - void fillSupplierProductTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void supProdInsertActionPerformed(java.awt.event.ActionEvent evt) - void supProdEditActionPerformed(java.awt.event.ActionEvent evt) - void supProdDeleteActionPerformed(java.awt.event.ActionEvent evt) - void supProdCancelActionPerformed(java.awt.event.ActionEvent evt) - void supplierProductTableMouseClicked(java.awt.event.MouseEvent evt) - void supProductSearchKeyReleased(java.awt.event.KeyEvent evt) - void supProdSupplierTableMouseClicked(java.awt.event.MouseEvent evt) - void supProdProductTableMouseClicked(java.awt.event.MouseEvent evt) - void supProdProductSearchActionPerformed(java.awt.event.ActionEvent evt) - void supProdProductSearchKeyReleased(java.awt.event.KeyEvent evt) - void supplierIdKeyReleased(java.awt.event.KeyEvent evt) - void productIdKeyReleased(java.awt.event.KeyEvent evt) - void itemIdCodeKeyPressed(java.awt.event.KeyEvent evt)

Brand
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - static Logger log - int brandid - javax.swing.JButton brandButCancel - javax.swing.JButton brandButDelete - javax.swing.JButton brandButEdit - javax.swing.JButton brandButInsert - app.bolivia.swing.JTextField brandCompanyNameAddress - app.bolivia.swing.JTextField brandCompanyName - app.bolivia.swing.JTextField brandCompanyRegNo - app.bolivia.swing.JTextField brandName - app.bolivia.swing.JTextField brandSearch - rojeru_san.complementos.RSTableMetro brandTable - javax.swing.JLabel jLabel6 - javax.swing.JPanel jPanel10 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane2
+ Brand() - boolean validateForm() + void clearBrandPanel() + void fillBrandTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void brandTableMouseClicked(java.awt.event.MouseEvent evt) - void brandSearchActionPerformed(java.awt.event.ActionEvent evt) - void brandSearchKeyReleased(java.awt.event.KeyEvent evt) - void brandButInsertActionPerformed(java.awt.event.ActionEvent evt) - void brandButEditActionPerformed(java.awt.event.ActionEvent evt) - void brandButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void brandButCancelActionPerformed(java.awt.event.ActionEvent evt) - void brandNameKeyReleased(java.awt.event.KeyEvent evt) - void brandNameKeyPressed(java.awt.event.KeyEvent evt) - void brandCompanyNameKeyPressed(java.awt.event.KeyEvent evt) - void brandCompanyAddressKeyPressed(java.awt.event.KeyEvent evt) - void brandCompanyRegNoKeyPressed(java.awt.event.KeyEvent evt)

Main
- static Logger log - Timer myTimer - TimerTask task - com.xpos.gui.ProductBatch Batch - com.xpos.gui.Brand Brand - com.xpos.gui.Category Category - com.xpos.gui.Customer Customer - com.xpos.gui.EmpPosition EmpPosition - com.xpos.gui.Employee Employee - com.xpos.gui.ExCost ExCost - com.xpos.gui.ExIncome ExIncome - javax.swing.JPanel HRM - javax.swing.JPanel Header - javax.swing.JPanel Invoices - javax.swing.JPanel MainPanel - com.xpos.gui.POIInvoice POIInvoice - com.xpos.gui.Product Product - javax.swing.JPanel ProductDetail - com.xpos.gui.Purchase Purchase - com.xpos.gui.PurchaseInvoice PurchaseInvoice - com.xpos.gui.PurchaseOrder PurchaseOrder - com.xpos.gui.QuateInvoice QuateInvoice - com.xpos.gui.Return Return - com.xpos.gui.ReturnInvoice ReturnInvoice - com.xpos.gui.Sale Sale - com.xpos.gui.SaleInvoice SaleInvoice - javax.swing.JPanel SideBar - com.xpos.gui.SupplierProduct SupplierProduct - com.xpos.gui.User User - com.xpos.gui.UserProfile UserProfile - javax.swing.JButton brandPanelBtn - com.xpos.dashboard.component.Dashpanel categoryCount - javax.swing.JButton categoryPanelBtn - javax.swing.JButton customer - com.xpos.dashboard.component.Dashpanel dailyCustomerCount - com.xpos.dashboard.component.Dashpanel dailyPurchaseCount - com.xpos.dashboard.component.Dashpanel dailyReturnCount - com.xpos.dashboard.component.Dashpanel dailySaleCount - javax.swing.JPanel dashboard - javax.swing.JButton employee - com.xpos.dashboard.component.Dashpanel employeeCount - javax.swing.JButton exCost - javax.swing.JButton exIncome - javax.swing.JButton homeBut - javax.swing.JButton hrm - javax.swing.JButton invoiceBtn - javax.swing.JLabel jLabel1 - javax.swing.JLabel jLabel45 - javax.swing.JLabel jLabel46 - javax.swing.JLabel jLabel47 - javax.swing.JPanel jPanel3 - javax.swing.JPanel jPanel35 - javax.swing.JPanel jPanel36 - javax.swing.JPanel jPanel37 - javax.swing.JPanel jPanel4 - com.xpos.dashboard.component.Dashpanel ongoingPOs - com.xpos.dashboard.component.Dashpanel ongoingQuates - javax.swing.JButton product - javax.swing.JButton productBatchPanelBtn - com.xpos.dashboard.component.Dashpanel productCount - javax.swing.JButton productPanelBtn - javax.swing.JButton purchase - javax.swing.JButton purchaseInvoiceBtn - javax.swing.JButton purchaseOrder - javax.swing.JButton purchaseOrderBtn - javax.swing.JButton quateRequestBtn - com.xpos.dashboard.component.Dashpanel remProductCount - com.xpos.gui.ReqForQuate reqForQuate - javax.swing.JButton reqForQuateBtn - javax.swing.JButton returnInvoiceBtn - javax.swing.JButton returnPanel - javax.swing.JButton sale - javax.swing.JButton saleInvoiceBtn - com.xpos.dashboard.component.Dashpanel stockItemCount - com.xpos.gui.Supply supplier - javax.swing.JButton supplierBut - com.xpos.dashboard.component.Dashpanel supplierCount - javax.swing.JButton supplierProduct - javax.swing.JLabel systemDate - javax.swing.JLabel systemTime - javax.swing.JLabel systemUserName - javax.swing.JButton userPosition - javax.swing.JButton userProfile
+ Main() + void runDateTime() + void setDashboardComponent() + void updateDashboard() + void setVisible falseAllPanels() - // <editor-fold defaultstate="collapsed" desc="Generated Code">#GEN-BEGIN: initComponents void initComponents() - void homeButActionPerformed(java.awt.event.ActionEvent evt) - void supplierButActionPerformed(java.awt.event.ActionEvent evt) - void productActionPerformed(java.awt.event.ActionEvent evt) - void customerActionPerformed(java.awt.event.ActionEvent evt) - void exCostActionPerformed(java.awt.event.ActionEvent evt) - void exIncomeActionPerformed(java.awt.event.ActionEvent evt) - void purchaseActionPerformed(java.awt.event.ActionEvent evt) - void saleActionPerformed(java.awt.event.ActionEvent evt) - void hrmActionPerformed(java.awt.event.ActionEvent evt) - void employeeActionPerformed(java.awt.event.ActionEvent evt) - void userProfileActionPerformed(java.awt.event.ActionEvent evt) - void userPositionActionPerformed(java.awt.event.ActionEvent evt) - void supplierProductActionPerformed(java.awt.event.ActionEvent evt) - void purchaseOrderActionPerformed(java.awt.event.ActionEvent evt) - void returnPanelActionPerformed(java.awt.event.ActionEvent evt) - void productBatchPanelBtnActionPerformed(java.awt.event.ActionEvent evt) - void categoryPanelBtnActionPerformed(java.awt.event.ActionEvent evt) - void brandPanelBtnActionPerformed(java.awt.event.ActionEvent evt) - void productPanelBtnActionPerformed(java.awt.event.ActionEvent evt) - void purchaseInvoiceBtnActionPerformed(java.awt.event.ActionEvent evt) - void purchaseOrderBtnActionPerformed(java.awt.event.ActionEvent evt) - void returnInvoiceBtnActionPerformed(java.awt.event.ActionEvent evt) - void saleInvoiceBtnActionPerformed(java.awt.event.ActionEvent evt) - void invoiceBtnActionPerformed(java.awt.event.ActionEvent evt) - void reqForQuateBtnActionPerformed(java.awt.event.ActionEvent evt) - void systemUserNameMouseClicked(java.awt.event.MouseEvent evt) - void quateRequestBtnActionPerformed(java.awt.event.ActionEvent evt) + static void main(String args)

Category
- DefaultTableModel tableModel - DateFormatter defaultDateFormat - static Logger log - int catId - javax.swing.JButton catButCancel - javax.swing.JButton catButDelete - javax.swing.JButton catButEdit - javax.swing.JButton catButInsert - app.bolivia.swing.JTextField catDescription - app.bolivia.swing.JTextField catSearch - app.bolivia.swing.JTextField catSpecNote - rojeru_san.complementos.RSTableMetro categoryTable - javax.swing.JLabel jLabel6 - javax.swing.JPanel jPanel10 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane2
+ Category() - boolean validateForm() + void clearCategoryPanel() - void fillCategoryTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code">#GEN-BEGIN: initComponents void initComponents() - void categoryTableMouseClicked(java.awt.event.MouseEvent evt) - void catSearchActionPerformed(java.awt.event.ActionEvent evt) - void catSearchKeyReleased(java.awt.event.KeyEvent evt) - void catButInsertActionPerformed(java.awt.event.ActionEvent evt) - void catButEditActionPerformed(java.awt.event.ActionEvent evt) - void catButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void catButCancelActionPerformed(java.awt.event.ActionEvent evt) - void catDescriptionKeyPressed(java.awt.event.KeyEvent evt)

Purchase
- DefaultTableModel tableModel - DateFormatter defaultDateFormat - DateFormatter defaultTimeFormat - Double totalWithNoDiscount - Double discount - Double finalTotal - int purchaseOrderId - int supplierId - javax.swing.JLabel jLabel13 - javax.swing.JLabel jLabel14 - javax.swing.JLabel jLabel40 - javax.swing.JLabel jLabel41 - javax.swing.JLabel jLabel42 - javax.swing.JPanel jPanel25 - javax.swing.JPanel jPanel26 - javax.swing.JPanel jPanel27 - javax.swing.JPanel jPanel3 - javax.swing.JScrollPane jScrollPane10 - javax.swing.JScrollPane jScrollPane11 - javax.swing.JScrollPane jScrollPane9 - javax.swing.JButton purchAddToTable - app.bolivia.swing.JTextField purchBarcode - javax.swing.JLabel purchBatchId - rojeru_san.complementos.RSTableMetro purchBatchTable - javax.swing.JButton purchCancel - javax.swing.JButton purchComplete - javax.swing.JButton purchDeleteItem - javax.swing.JLabel purchDescription - javax.swing.JRadioButton purchDisAmount - javax.swing.ButtonGroup purchDisGroup - javax.swing.JRadioButton purchDisPercent - app.bolivia.swing.JTextField purchDiscount - javax.swing.JLabel purchExpireDate - javax.swing.JLabel purchFinalDiscount - javax.swing.JLabel purchManufacDate - app.bolivia.swing.JTextField purchProdBarcode - app.bolivia.swing.JTextField purchProdBatchNumber - app.bolivia.swing.JTextField purchProdDescription - rojeru_san.complementos.RSTableMetro purchProdTable - rojeru_san.complementos.RSTableMetro purchProduct - app.bolivia.swing.JTextField purchProductId - javax.swing.JComboBox<String> purchPurchaseOrderCombo - app.bolivia.swing.JTextField purchPurchasePrice - app.bolivia.swing.JTextField purchQuantity - javax.swing.JLabel purchSubTotal - javax.swing.JComboBox<String> purchSupplierCombo - javax.swing.JLabel purchTotal
+ Purchase() - // <editor-fold defaultstate="collapsed" desc="Generated Code">#GEN-BEGIN: initComponents void initComponents() - void loadPurchaseSupplierCombo() - void loadPurchaseOrderCombo() - void fillPurchaseProdTable(String query) - void fillPurchaseBatchTable(String query) + void calPurchaseDetails() + void clearPurchPanel() - void purchProdTableMouseClicked(java.awt.event.MouseEvent evt) - void purchProdDescriptionKeyReleased(java.awt.event.KeyEvent evt) - void purchBatchTableMouseClicked(java.awt.event.MouseEvent evt) - void purchProdBatchNumberKeyReleased(java.awt.event.KeyEvent evt) - void purchProductIdKeyReleased(java.awt.event.KeyEvent evt) - void purchBarcodeKeyReleased(java.awt.event.KeyEvent evt) - void purchQuantityKeyReleased(java.awt.event.KeyEvent evt) - void purchPurchasePriceKeyReleased(java.awt.event.KeyEvent evt) - void purchAddToTableActionPerformed(java.awt.event.ActionEvent evt) - void purchCompleteActionPerformed(java.awt.event.ActionEvent evt) - void printInvoicePrintInvoice() - void purchDeleteItemActionPerformed(java.awt.event.ActionEvent evt) - void purchCancelActionPerformed(java.awt.event.ActionEvent evt) - void purchDiscountKeyReleased(java.awt.event.KeyEvent evt) - void purchDisAmountMouseClicked(java.awt.event.MouseEvent evt) - void purchDisPercentMouseClicked(java.awt.event.MouseEvent evt) - void purchProductMouseClicked(java.awt.event.MouseEvent evt) - void purchSupplierComboActionPerformed(java.awt.event.ActionEvent evt) - void purchProdBarcodeKeyReleased(java.awt.event.KeyEvent evt) - void purchPurchaseOrderComboActionPerformed(java.awt.event.ActionEvent evt)

SaleInvoice
- javax.swing.JButton btnInvoicePrint - javax.swing.JLabel jLabel6 - javax.swing.JPanel jPanel11 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane2 - javax.swing.JScrollPane jScrollPane3 - app.bolivia.swing.JTextField xCstTextField saleSearchByCustomer - app.bolivia.swing.JTextField saleSearchByInvoice - rojeru_san.complementos.RSTableMetro saleTable - rojeru_san.complementos.RSTableMetro soldItemTable
<pre> + SaleInvoice() + void fillInvoiceTable(String query) + void fillSoldItemTable(int saleId) + // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() + void saleTableMouseClicked(java.awt.event.MouseEvent evt) + void saleSearchByInvoiceActionPerformed(java.awt.event.ActionEvent evt) + void saleSearchByInvoiceKeyReleased(java.awt.event.KeyEvent evt) + void soldItemTableMouseClicked(java.awt.event.MouseEvent evt) + void saleSearchByCustomerActionPerformed(java.awt.event.ActionEvent evt) + void saleSearchByCustomerKeyReleased(java.awt.event.KeyEvent evt) + void btnInvoicePrintActionPerformed(java.awt.event.ActionEvent evt) </pre>

ExCost
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - static Logger log - int xCostId - javax.swing.JLabel jLabel11 - javax.swing.JPanel jPanel21 - javax.swing.JPanel jPanel23 - javax.swing.JScrollPane jScrollPane7 - app.bolivia.swing.JTextField xCstAmount - javax.swing.JButton xCstButCancel - javax.swing.JButton xCstButDelete - javax.swing.JButton xCstButEdit - javax.swing.JButton xCstButInsert - com.toedter.calendar.JDateChooser xCstDate - app.bolivia.swing.JTextField xCstDescription - app.bolivia.swing.JTextField xCstSearch - rojeru_san.complementos.RSTableMetro xCstTable
<pre> + ExCost() + boolean validateForm() + void clearExCostPanel() + void fillExCostTable(String query) + // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() + void xCstButInsertActionPerformed(java.awt.event.ActionEvent evt) + void xCstButEditActionPerformed(java.awt.event.ActionEvent evt) + void xCstButDeleteActionPerformed(java.awt.event.ActionEvent evt) + void xCstButCancelActionPerformed(java.awt.event.ActionEvent evt) + void xCstTableMouseClicked(java.awt.event.MouseEvent evt) + void xCstSearchKeyReleased(java.awt.event.KeyEvent evt) + void xCstDateKeyPressed(java.awt.event.KeyEvent evt) + void xCstDescriptionKeyPressed(java.awt.event.KeyEvent evt) + void xCstAmountKeyPressed(java.awt.event.KeyEvent evt) </pre>

ReqForQuate
- int supplierId - DefaultTableModel tablemodel - DateFormatter defaultDateFormat - DateFormatter defaultDateFormat - javax.swing.JButton RFQAddToTable - javax.swing.JLabel RFQBrand - javax.swing.JButton RFQCancel - javax.swing.JButton RFQComplete - javax.swing.JButton RFQDeleteItem - javax.swing.JLabel RFQDescription - javax.swing.JLabel RFQItemCode - javax.swing.JLabel RFQOrderedQty - app.bolivia.swing.JTextField RFQProdDescriptionSearch - javax.swing.JLabel RFQProductId - app.bolivia.swing.JTextField RFQQuantity - javax.swing.JLabel RFQStock - javax.swing.JComboBox<String> RFQSupplierCombo - rojeru_san.complementos.RSTableMetro RFQSupplierProduct - rojeru_san.complementos.RSTableMetro RFQTable - javax.swing.JLabel jLabel13 - javax.swing.JPanel jPanel25 - javax.swing.JPanel jPanel26 - javax.swing.JPanel jPanel3 - javax.swing.JPanel jPanel5 - javax.swing.JScrollPane jScrollPane11 - javax.swing.JScrollPane jScrollPane9
<pre> + ReqForQuate() + void clearPurchOrderPanel() + void loadPurchaseOrderSupplierCombo() + void fillQuateReqProductTable(int suplid) + void fillSupplierProductTable(String query) + // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() + void RFQSupplierProductMouseClicked(java.awt.event.MouseEvent evt) + void RFQProdDescriptionSearchKeyReleased(java.awt.event.KeyEvent evt) + void RFQQuantityKeyReleased(java.awt.event.KeyEvent evt) + void RFQAddToTableActionPerformed(java.awt.event.ActionEvent evt) + void RFQCompleteActionPerformed(java.awt.event.ActionEvent evt) + void printInvoice(int invoiceId) + void RFQDeleteItemActionPerformed(java.awt.event.ActionEvent evt) + void RFQCancelActionPerformed(java.awt.event.ActionEvent evt) + void RFQTableMouseClicked(java.awt.event.MouseEvent evt) + void RFQSupplierComboActionPerformed(java.awt.event.ActionEvent evt) </pre>

EmpPosition
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - static Logger log - int empPosId - javax.swing.JButton empPosButCancel - javax.swing.JButton empPosButDelete - javax.swing.JButton empPosButEdit - javax.swing.JButton empPosButInsert - app.bolivia.swing.JTextField empPosName - javax.swing.JCheckBox empPosPerCustomer - javax.swing.JCheckBox empPosPerEmployee - javax.swing.JCheckBox empPosPerInvoice - javax.swing.JCheckBox empPosPerProducts - javax.swing.JCheckBox empPosPerPurchase - javax.swing.JCheckBox empPosPerReport - javax.swing.JCheckBox empPosPerReturn - javax.swing.JCheckBox empPosPerSale - javax.swing.JCheckBox empPosPerSupplier - javax.swing.JCheckBox empPosPerUserPosition - javax.swing.JCheckBox empPosPerUserProfile - javax.swing.JCheckBox empPosPerXCost - javax.swing.JCheckBox empPosPerXIncome - app.bolivia.swing.JTextField empPosSearch - rojeru_san.complementos.RSTableMetro empPosTable - javax.swing.JLabel jLabel8 - javax.swing.JPanel jPanel13 - javax.swing.JPanel jPanel14 - javax.swing.JPanel jPanel2 - javax.swing.JScrollPane jScrollPane4
<pre> + EmpPosition() + void clearEmpPosPanel() + void fillEmpPosTable(String query) + // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() + void empPosButInsertActionPerformed(java.awt.event.ActionEvent evt) + void empPosButEditActionPerformed(java.awt.event.ActionEvent evt) + void empPosButDeleteActionPerformed(java.awt.event.ActionEvent evt) + void empPosButCancelActionPerformed(java.awt.event.ActionEvent evt) + void empPosPerProductsActionPerformed(java.awt.event.ActionEvent evt) + void empPosTableMouseClicked(java.awt.event.MouseEvent evt) + void empPosSearchKeyReleased(java.awt.event.KeyEvent evt) </pre>

Employee
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - static Logger log - int employeeid - app.bolivia.swing.JTextField empAddress - javax.swing.JButton empButCancel - javax.swing.JButton empButDelete - javax.swing.JButton empButEdit - javax.swing.JButton empButInsert - com.toedter.calendar.JDateChooser empDOB - app.bolivia.swing.JTextField empEmail - javax.swing.JRadioButton empGenFemale - javax.swing.JRadioButton empGenMale - com.toedter.calendar.JDateChooser empJoinedDate - app.bolivia.swing.JTextField empNIC - app.bolivia.swing.JTextField empName - javax.swing.JComboBox<String> empPositionCombo - app.bolivia.swing.JTextField empSearch - rojeru_san.complementos.RSTableMetro empTable - app.bolivia.swing.JTextField empTeleNumber - javax.swing.ButtonGroup gender - javax.swing.JLabel jLabel9 - javax.swing.JPanel jPanel15 - javax.swing.JPanel jPanel16 - javax.swing.JPanel jPanel17 - javax.swing.JScrollPane jScrollPane5
+ Employee() + void clearEmployeePanel() - boolean validateForm() + void loadEmpPosCombo() + void fillEmpTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void empButInsertActionPerformed(java.awt.event.ActionEvent evt) - void empButEditActionPerformed(java.awt.event.ActionEvent evt) - void empButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void empButCancelActionPerformed(java.awt.event.ActionEvent evt) - void empTableMouseClicked(java.awt.event.MouseEvent evt) - void empSearchKeyReleased(java.awt.event.KeyEvent evt) - void empNameKeyPressed(java.awt.event.KeyEvent evt) - void empAddressKeyPressed(java.awt.event.KeyEvent evt) - void empTeleNumberKeyPressed(java.awt.event.KeyEvent evt) - void empNICKeyPressed(java.awt.event.KeyEvent evt) - void empDOBKeyPressed(java.awt.event.KeyEvent evt) - void empJoinedDateKeyPressed(java.awt.event.KeyEvent evt) - void empEmailKeyPressed(java.awt.event.KeyEvent evt) - void empGenMaleKeyPressed(java.awt.event.KeyEvent evt) - void empGenFemaleKeyPressed(java.awt.event.KeyEvent evt) - void empPositionComboKeyPressed(java.awt.event.KeyEvent evt)

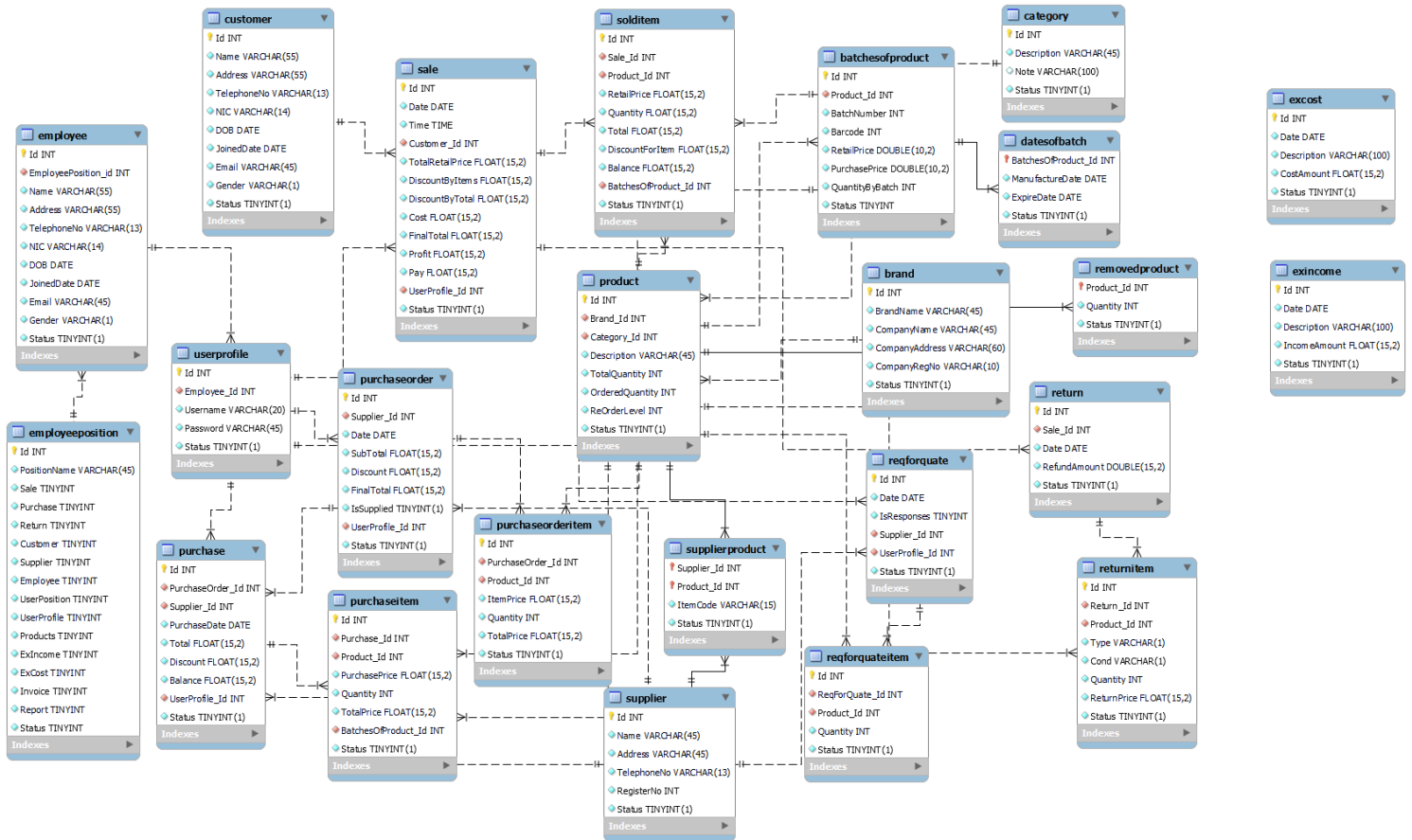
ReturnInvoice
- javax.swing.JButton buttInvoicePrint - javax.swing.JLabel jLabel6 - javax.swing.JPanel jPanel11 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane2 - javax.swing.JScrollPane jScrollPane3 - rojeru_san.complementos.RSTableMetro returnItemTable - app.bolivia.swing.JTextField returnSearchByCustomer - app.bolivia.swing.JTextField returnSearchByReturnInvoice - app.bolivia.swing.JTextField returnSearchBySaleInvoice - rojeru_san.complementos.RSTableMetro returnTable
+ ReturnInvoice() + void fillReturnInvoiceTable(String query) - void fillReturnItemTable(int returnid) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void returnTableMouseClicked(java.awt.event.MouseEvent evt) - void returnSearchBySaleInvoiceActionPerformed(java.awt.event.ActionEvent evt) - void returnSearchBySaleInvoiceKeyReleased(java.awt.event.KeyEvent evt) - void returnItemTableMouseClicked(java.awt.event.MouseEvent evt) - void returnSearchByCustomerActionPerformed(java.awt.event.ActionEvent evt) - void returnSearchByCustomerKeyReleased(java.awt.event.KeyEvent evt) - void returnSearchByReturnInvoiceActionPerformed(java.awt.event.ActionEvent evt) - void returnSearchByReturnInvoiceKeyReleased(java.awt.event.KeyEvent evt) - void buttInvoicePrintActionPerformed(java.awt.event.ActionEvent evt)

SignIn
- static Logger log - javax.swing.ButtonGroup buttonGroup1 - javax.swing.JLabel jLabel1 - javax.swing.JLabel jLabel2 - javax.swing.JPanel jPanel1 - javax.swing.JPanel jPanel2 - javax.swing.JButton login - com.xpox.custom.JCPasswordField password - app.bolivia.swing.JTextField username
+ SignIn() + void signInProcess() - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void usernameKeyReleased(java.awt.event.KeyEvent evt) - void loginActionPerformed(java.awt.event.ActionEvent evt) - void passwordKeyPressed(java.awt.event.KeyEvent evt) + static void main(String args)

Product
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - int productid - javax.swing.JLabel jLabel6 - javax.swing.JPanel jPanel10 - javax.swing.JPanel jPanel9 - javax.swing.JScrollPane jScrollPane2 - javax.swing.JComboBox<String> prodBrandCombo - javax.swing.JButton prodButCancel - javax.swing.JButton prodButDelete - javax.swing.JButton prodButEdit - javax.swing.JButton prodButInsert - javax.swing.JComboBox<String> prodCategoryCombo - app.bolivia.swing.JTextField prodDescription - app.bolivia.swing.JTextField prodOrderedQuantity - app.bolivia.swing.JTextField prodQuantity - app.bolivia.swing.JTextField prodReOrderLevel - app.bolivia.swing.JTextField prodSearch - rojeru_san.complementos.RSTableMetro productTable
+ Product() - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - boolean validateForm() + void loadProdBrandCombo() + void loadProdCategoryCombo() + void fillProductTable(String query) + void clearProductPanel() - void productTableMouseClicked(java.awt.event.MouseEvent evt) - void prodSearchActionPerformed(java.awt.event.ActionEvent evt) - void prodSearchKeyReleased(java.awt.event.KeyEvent evt) - void prodButInsertActionPerformed(java.awt.event.ActionEvent evt) - void prodButEditActionPerformed(java.awt.event.ActionEvent evt) - void prodButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void prodButCancelActionPerformed(java.awt.event.ActionEvent evt) - void prodDescriptionKeyPressed(java.awt.event.KeyEvent evt) - void prodBrandComboKeyPressed(java.awt.event.KeyEvent evt) - void prodCategoryComboKeyPressed(java.awt.event.KeyEvent evt) - void prodReOrderLevelKeyPressed(java.awt.event.KeyEvent evt) - void prodOrderedQuantityKeyPressed(java.awt.event.KeyEvent evt) - void prodQuantityKeyPressed(java.awt.event.KeyEvent evt)

UserProfile
- DefaultTableModel tablemodel - DateFormatter defaultDateFormat - int userProid - javax.swing.ButtonGroup isActive - javax.swing.JLabel jLabel10 - javax.swing.JPanel jPanel18 - javax.swing.JPanel jPanel19 - javax.swing.JPanel jPanel20 - javax.swing.JScrollPane jScrollPane6 - javax.swing.JRadioButton userProIsActive - javax.swing.JButton userProButCancel - javax.swing.JButton userProButDelete - javax.swing.JButton userProButEdit - javax.swing.JButton userProButInsert - javax.swing.JComboBox<String> userProEmployeeCombo - javax.swing.JRadioButton userProIsActive - javax.swing.JPasswordField userProPassword - app.bolivia.swing.JTextField userProSearch - rojeru_san.complementos.RSTableMetro userProTable - app.bolivia.swing.JTextField userProUsername
+ UserProfile() + void clearUserProPanel() + void loadUserProCombo() + void fillUserProTable(String query) - // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents void initComponents() - void userProButInsertActionPerformed(java.awt.event.ActionEvent evt) - void userProButEditActionPerformed(java.awt.event.ActionEvent evt) - void userProButDeleteActionPerformed(java.awt.event.ActionEvent evt) - void userProButCancelActionPerformed(java.awt.event.ActionEvent evt) - void userProTableMouseClicked(java.awt.event.MouseEvent evt) - void userProSearchKeyReleased(java.awt.event.KeyEvent evt)

EER Diagram



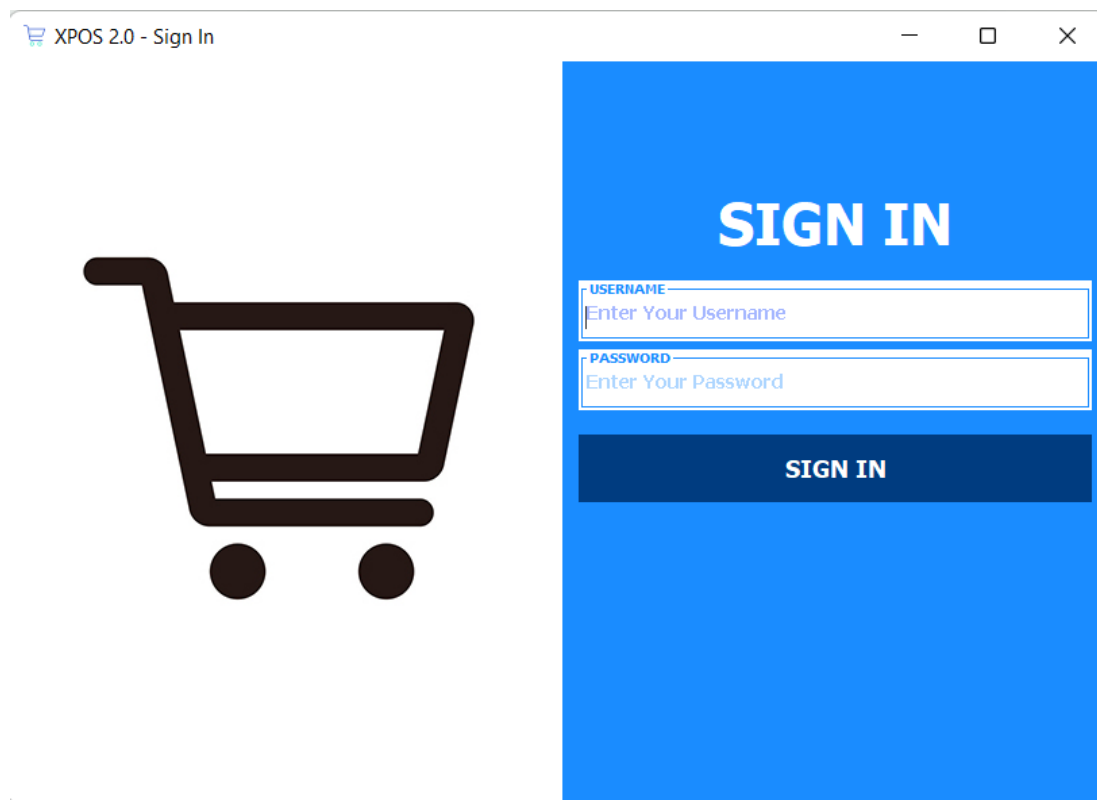
Specific Requirements

External Interface Requirements

1. User Interfaces

Sign In Interface

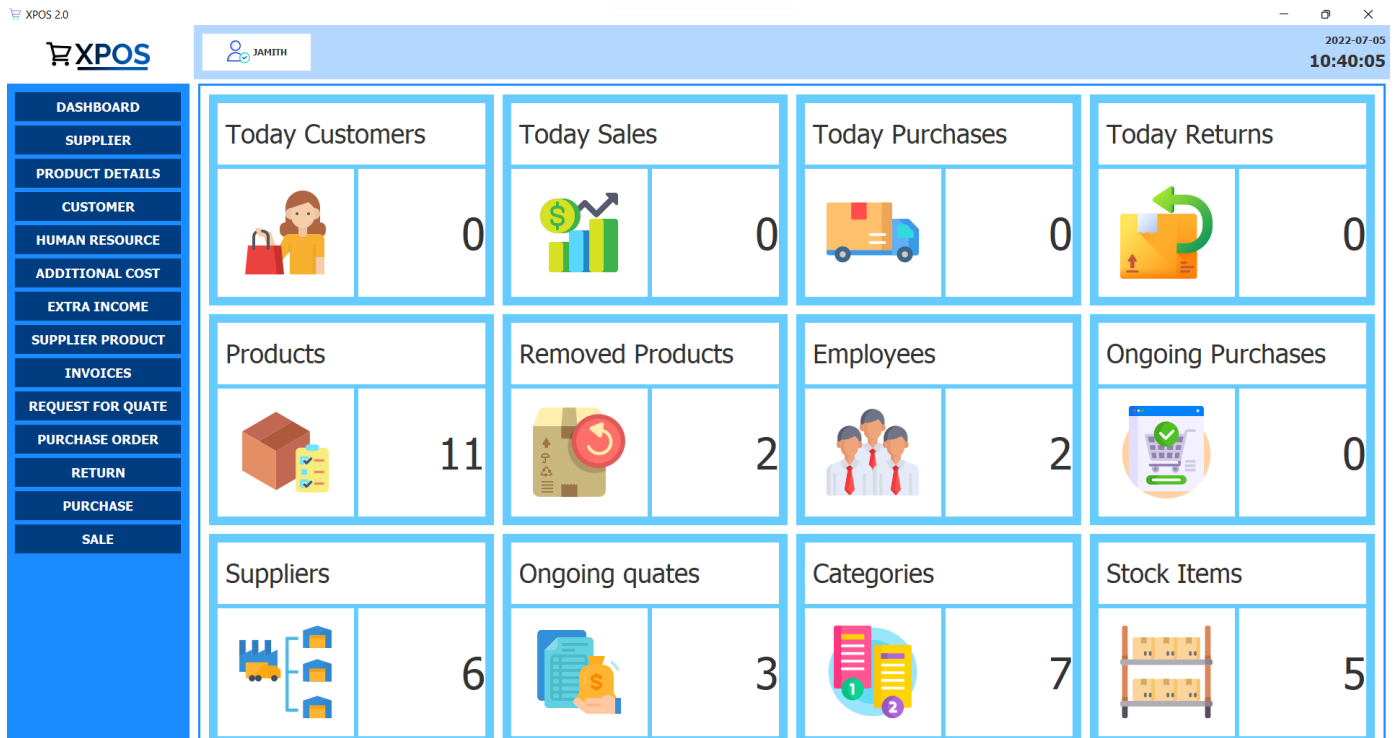
All the users will see the login page when they enter the system. This page asks the users a username and a password which will provide the authorization to enter the system



Dashboard Interface

The dashboard can be accessed from the main panel. The main panel always checks to see if the person who is signed in has access to something. If there is no access, a message is sent.

In addition, the dashboard is updated in real-time.



User Interface

The user who is using the system at that time is shown above. There the user has the ability to update his details.

The screenshot shows the 'USER' profile page in the XPOS 2.0 system. The top navigation bar is identical to the dashboard. The left sidebar menu is also present. The main content area is titled 'USER' and contains a form for updating user details. The form has three input fields for 'Username', 'New Password', and 'Confirm Password'. Below these fields are 'UPDATE' and 'CANCEL' buttons. To the right of the form, a list of user details is displayed:

Employee ID	3
Employee Position	admin
Name	jamith
Address	nainamadama
Telephone No	0770470323
NIC	200022502886
Date Of Birth	2022-06-09
Joined Date	2022-06-03
Email	jamithsheron@gmail.com
Gender	M

Supplier Interface

The data of the suppliers who provide the goods to the shop is entered through this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

Product Interface

The details of the products released by each company's brands are added to the system. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

Product Batches Interface

The details of stock are managed here. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

XPOS 2.0

XPOS

JAMITHI

2022-07-05 04:37:02

PRODUCT DETAILS

PRODUCT BATCHES

Product Id:

Description:

Barcode:

Batch Number:

Quantity:

Purchase Price:

Retail Price:

☐ Include Below Dates

Manufacture Date:

Expire Date:

Batch Details

Search:

ID	Barcode	Product Desc...	Batch Number	Quantity	Purchase Price	Retail Price	Manufacture Date	Expire Date
7	123456987	6 - Square rule	985	83	65.0	100.0		
8	321654984	7 - single rule	986	48	58.96	103.0		
9	654321456	8 - choctet ice cr...	145	0	50.0	70.0	2021-07-15	2023-07-15
10	321654984	9 - vanilla ice cre...	124	0	0.0	75.0	2021-07-09	2023-07-13
11	654354985	13 - choctet milk	321	113	55.83	70.0	2022-03-10	2022-12-20
12	654907254	14 - vanilla milk	564	126	62.94	70.0	2022-03-16	2022-12-14
13	654324953	12 - choctet	231	100	55.0	70.0	2022-07-06	2023-07-19

Product Details

Search:

ID	Description	Brand	Category	Re-Order Level	Ordered Quantity	Quantity
6	Square rule	5 - Atlas	4 - School Item	100	0	183
7	single rule	5 - Atlas	4 - School Item	100	0	48
8	choctet ice cream	7 - Ice cream elephan...	5 - Ice cream	20	0	0
9	vanilla ice cream	7 - Ice cream elephan...	5 - Ice cream	20	0	0
10	coca cola	6 - coca cola	7 - soft drinks	10	0	0
11	sprite	6 - coca cola	7 - soft drinks	10	0	0
12	choctet	9 - kandos choctet	6 - choctet	40	0	100
13	choctet milk	8 - kothmale fresh milk	8 - milk	15	0	114
14	vanilla milk	8 - kothmale fresh milk	8 - milk	10	0	130

BRAND

Brand Name:

Company Name:

Company Address:

Company Register NO:

BRAND

Search:

ID	Brand	Company	Company Address	Company Reg NO
5	Atlas	Atlas Stationaries	Colombo 11.	654654
6	coca cola	coca cola	colombo	752145
7	Ice cream elephant house	elephant house	colombo	852340
8	kothmale fresh milk	kothmale	rajama	023564
9	kandos choctet	kandos	rajama	526879
10	maliban bilcat	maliban	rathmalana	025312
11	mades chikan	mades	panadura	325641

PRODUCT DETAILS

PRODUCT **PRODUCT BATCH** **BRAND** **CATEGORY**

BRAND

Brand Name:

Company Name:

Company Address:

Company Register NO:

BRAND

Search:

ID	Brand	Company	Company Address	Company Reg NO
5	Atlas	Atlas Stationaries	Colombo 11.	654654
6	coca cola	coca cola	colombo	752145
7	Ice cream elephant house	elephant house	colombo	852340
8	kothmale fresh milk	kothmale	rajama	023564
9	kandos choctet	kandos	rajama	526879
10	maliban bilcat	maliban	rathmalana	025312
11	mades chikan	mades	panadura	325641

PRODUCT DETAILS

PRODUCT **PRODUCT BATCH** **BRAND** **CATEGORY**

BRAND

Brand Name:

Company Name:

Company Address:

Company Register NO:

BRAND

Search:

ID	Brand	Company	Company Address	Company Reg NO
5	Atlas	Atlas Stationaries	Colombo 11.	654654
6	coca cola	coca cola	colombo	752145
7	Ice cream elephant house	elephant house	colombo	852340
8	kothmale fresh milk	kothmale	rajama	023564
9	kandos choctet	kandos	rajama	526879
10	maliban bilcat	maliban	rathmalana	025312
11	mades chikan	mades	panadura	325641

Product Brand Interface

Brands related to products details are managed through this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

XPOS 2.0

XPOS

JAMITHI

2022-07-05 04:43:27

PRODUCT DETAILS

BRAND

Brand Name:

Company Name:

Company Address:

Company Register NO:

BRAND

Search:

ID	Brand	Company	Company Address	Company Reg NO
5	Atlas	Atlas Stationaries	Colombo 11.	654654
6	coca cola	coca cola	colombo	752145
7	Ice cream elephant house	elephant house	colombo	852340
8	kothmale fresh milk	kothmale	rajama	023564
9	kandos choctet	kandos	rajama	526879
10	maliban bilcat	maliban	rathmalana	025312
11	mades chikan	mades	panadura	325641

PRODUCT DETAILS

PRODUCT **PRODUCT BATCH** **BRAND** **CATEGORY**

BRAND

Brand Name:

Company Name:

Company Address:

Company Register NO:

BRAND

Search:

ID	Brand	Company	Company Address	Company Reg NO
5	Atlas	Atlas Stationaries	Colombo 11.	654654
6	coca cola	coca cola	colombo	752145
7	Ice cream elephant house	elephant house	colombo	852340
8	kothmale fresh milk	kothmale	rajama	023564
9	kandos choctet	kandos	rajama	526879
10	maliban bilcat	maliban	rathmalana	025312
11	mades chikan	mades	panadura	325641

Product Category Interface

The category to which the relevant product belongs is managed here. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

ID	Description	Special Note
4	School Item	
5	Ice cream	store below at 16c
6	chocket	
7	soft drinks	
8	milk	
9	biscuits	
10	meat	

Customer Interface

Customer's details are managed through this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

ID	Name	Address	Telephone No	NIC	Date of Birth	Joined Date	Email	Gender
1	Temporary	address	0111111111	000000000000	1970-07-01	1970-07-01	mail@mail.com	Male
4	ashan	nainadadimbakjui	0776546542	200032132165	2022-06-16	2022-06-09	ashbasd@gmail.c...	Male

Employee Interface

Employees' details are managed through this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

EMPLOYEE

Name:

Address:

Telephone Number:

NIC:

Date Of Birth:

Joined Date:

Email:

Gender: ☒ Male ☐ Female

Select Position:

EMPLOYEE

ID	Name	Address	Telephone No	NIC	Date Of Birth	Joined Date	Email	Gender	Position ID	Position
3	jamith	nainamadams	0770470323	2000225028...	2022-06-09	2022-06-03	jamithshero...	Male	3	admin
4	kaveesha sh...	nainamadams	0763496745	993213721V	1999-11-06	2022-06-29	jamithshero...	Male	4	assistant

INSERT **EDIT** **DELETE** **CANCEL**

Employee Position Interface

Things that the employee can access in the system and things that cannot be accessed are managed through this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

EMP POSITIONS

Name:

Access Permission:

- ☐ Sale
- ☐ Purchase
- ☐ Return
- ☐ Customer
- ☐ Supplier
- ☐ Employee
- ☐ User Position
- ☐ User Profile
- ☐ Products
- ☐ Invoice
- ☐ Extra Income
- ☐ Extra Cost
- ☐ Report

EMP POSITIONS

ID	Name	Sale	Purchase	Return	Customer	Supplier	Employee	User Pos...	User Pro...	Products	Invoice	Extra In...	Extra Cost	Report
3	admin	Allow	Allow	Allow	Allow	Allow	Allow	Allow	Allow	Allow	Allow	Allow	Allow	Allow
4	assistant	Allow	Deny	Deny	Deny	Deny	Deny	Deny	Deny	Deny	Deny	Deny	Deny	Deny

INSERT **EDIT** **DELETE** **CANCEL**

User Profile Interface

If existing employees want to use the system, this interface will provide access and create an account by entering a username and password. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

ID	Employee Id	Employee Name	UserName	Status
2	3	janith	janith	Active
3	4	kaveesha shavinda	kavi	Not Active

Additional Cost Interface

Additional expenses are managed from this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

ID	Date	Description	Cost Amount
1	2022-06-09	green fund	5000.0

Extra Income Interface

Extra costs are managed from this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

EXTRA INCOME

Date: Search:

Description:

Amount:

INSERT EDIT DELETE CANCEL

ID	Date	Description	Income Amount
1	2022-06-16	electricity bill	2500.0

Supplier Products Interface

Some suppliers have codes for the convenience of identifying each product separately. Managing the product code is done through this interface. The data relating to the tablet is filled. There are functions to insert, edit, delete, and cancel. There is also the ability to search for data from the table.

SUPPLIER PRODUCTS

Supplier ID: Search:

Supplier Name:

Product ID:

Product Name:

Item Code:

INSERT EDIT DELETE CANCEL

Supplier ID	Supplier Name	Product ID	Product Name	Item Code
4	atlas supplier	6	Square rule	atsep
4	atlas supplier	7	single rule	atse
5	biyagama supplier	7	single rule	blse
5	biyagama supplier	16	chikan	ch011
12	elephant house ice cream	8	choclet ice cream	ch020
12	elephant house ice cream	9	vanilla ice cream	ch001
14	coka cola	10	coka cola	520
14	coka cola	11	sprite	521
15	kandos	12	choclet	ka585
16	kothmale	13	choclet milk	644
16	kothmale	14	vanila milk	445

ID	Name	Address	Telephone No	Register No
4	atlas supplier	colombo	0773216545	645654
5	biyagama supplier	rattmalana	0316546548	321654
12	elephant house ice cream	colombo	0112534652	125369
14	coka cola	colombo	0125964851	182456
15	kandos	rugama	0756542351	185672
16	kothmale	colombo	0778523645	913257

ID	Description	Brand	Category	Re-Order Level	Ordered Quantity	Quantity
6	Square rule	5 - Atlas	4 - School Item	100	0	183
7	single rule	5 - Atlas	4 - School Item	100	0	48
8	choclet ice cream	7 - ice cream	5 - ice cream	20	0	0
9	vanilla ice cream	7 - ice cream	5 - ice cream	20	0	0
10	coka cola	6 - coca cola	7 - soft drinks	10	0	0
11	sprite	6 - coca cola	7 - soft drinks	10	0	0
12	choclet	9 - kandos ch...	6 - choclet	40	0	100
13	choclet milk	8 - kothmale L...	8 - milk	15	0	114
14	vanila milk	8 - kothmale L...	8 - milk	10	0	129

Invoice Interface

Invoices of previous sales or transactions can be previewed from this interface.

➤ Sales invoices

The screenshot shows the XPOS 2.0 software interface. The sidebar on the left contains navigation links: DASHBOARD, SUPPLIER, PRODUCT DETAILS, CUSTOMER, HUMAN RESOURCE, ADDITIONAL COST, EXTRA INCOME, SUPPLIER PRODUCT, INVOICES, REQUEST FOR QUATE, PURCHASE ORDER, RETURN, PURCHASE, and SALE. The main area is titled 'INVOICES' and includes sub-sections for SALE INVOICE, PURCHASE INVOICE, RETURN INVOICE, QUATE REQUEST, and PURCHASE ORDER. The 'SALES INVOICES' section is active, displaying a table with the following data:

ID	Date - Time	Customer	Total Retail	Discount For	Grand Total	Pay	System User
11	2022-07-0...	4 - ashan	206.0	6.0	200.0	0.0	janith
12	2022-07-0...	4 - ashan	200.0	20.0	180.0	200.0	janith
13	2022-07-0...	4 - ashan	200.0	0.0	200.0	500.0	janith
14	2022-07-0...	1 - Tempora...	200.0	0.0	190.0	200.0	janith
15	2022-07-0...	1 - Tempora...	280.0	6.0	274.0	500.0	janith
16	2022-07-0...	1 - Tempora...	1630.0	55.0	1574.4	2000.0	janith
17	2022-07-0...	1 - Tempora...	100.0	0.0	100.0	100.0	janith

A 'PRINT INVOICE' button is located in the top right corner of the main area.

Here there are two invoices for sales invoices, that is, invoices that can be taken from an 80mm printer and an A4 printer.

A desired invoice can be obtained according to the need.

XPOS
11 Sea Street,
Colombo
0785423695
xpos@email.com

Sale Invoice

Sale No: 13
Date & Time: 7/1/22 7:54 AM

Description	Quantity	Price	Sub Total	Discount	Total
Square rule	2.00	100.00	200.00	0.00	200.00
Total		200.00	0.00		200.00
Dis. For				0.00	
Pay					500.00
Balance					300.00

Thank You For Your Business

XPOS
11 Sea Street,
Colombo
0785423695
xpos@email.com

Sale Invoice

Sale No: 13
Date & Time: July 1, 2022 7:54 AM

Product Description	Retail Price	Quantity	Sub Total	Discount	Total
Square rule	100.00	2.00	200.00	0.00	200.00

➤ Purchase invoices

XPOS 2.0

XPOS

2022-07-05 05:43:51

INVOICES

SALE INVOICE PURCHASE INVOICE RETURN INVOICE QUATE REQUEST PURCHASE ORDER

PURCHASE INVOICES

Invoice Number: Search Invoice Number Search Supplier

PRINT INVOICE

ID	Purchase Ord...	Supplier	Date	Total	Discount	GrandTotal	System User
3	7	4 - atlas supp...	2022-07-01	6500.0	0.0	6500.0	janith
4	8	16 - koffiniale	2022-07-02	11600.0	2320.0	9280.0	janith
5	9	15 - kandos	2022-07-02	5500.0	0.0	5500.0	janith
6	10	16 - koffiniale	2022-07-02	3475.0	0.0	3475.0	janith
7	11	4 - atlas supp...	2022-07-02	3000.0	0.0	3000.0	janith

ID - Product	ID - Batch Number	Purchase Price	Quantity	Total
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XPOS
11 Sea Street,
Colombo
0785423695
xpos@email.com

Good Receive Note

Purchase No: 5 Date : 7/2/22 12:00 AM

From:
Name kandos
Address ragam
Tele. No 0756542351
Reg. No 185672

Comments:

Quantity	Description	Unit Price	Total
100	choclet	55.00	5,500.00

➤ Return invoices

XPOS 2.0

XPOS

2022-07-05 05:45:34

INVOICES

SALE INVOICE PURCHASE INVOICE RETURN INVOICE QUATE REQUEST PURCHASE ORDER

RETURN INVOICES

Sale Invoice Number
 Return Invoice Number
 Search Customer

Search Sale Invoice No
 Search Return Invoice No
 Search Customer

PRINT INVOICE

ID	sale ID	Date	Customer	Total Refund Amount
9	11	2022-07-01	4 - ashan	100.0
10	12	2022-07-01	4 - ashan	90.0
11	13	2022-07-01	4 - ashan	100.0
12	16	2022-07-02	1 - Temporary	70.0

ID - Product	Return Type	Product Condition	Quantity	Refund Total

XPOS
11 Sea Street,
Colombo
0785423695
xpos@email.com

Return Invoice

Return No: 11 Sale No: 13 Date : July 1, 2022

Product Description	Return Type	Product Condition	Quantity	Return Price
Square rule	Cash Refund	Damage	1	100.00

XPOS 2.0

JAMITH

2022-07-05
05:59:30

DASHBOARD

SUPPLIER

PRODUCT DETAILS

CUSTOMER

HUMAN RESOURCE

ADDITIONAL COST

EXTRA INCOME

SUPPLIER PRODUCT

INVOICES

REQUEST FOR QUATE

PURCHASE ORDER

RETURN

PURCHASE

SALE

INVOICES

SALE INVOICE

PURCHASE INVOICE

RETURN INVOICE

QUATE REQUEST

PURCHASE ORDER

QUATE INVOICE

Invoice Number

Search Customer

SET RESPONDED

PRINT INVOICE

ID	Date	Supplier	Is Responed	System ther
4	2022-06-26	4 - atlas supplier	Responed	jamith
5	2022-06-26	4 - atlas supplier	Responed	jamith
6	2022-06-26	16 - kothmale	Responed	jamith
7	2022-07-01	16 - kothmale	Responed	jamith
8	2022-07-02	15 - handow	Not Responed	jamith
9	2022-07-02	15 - handow	Not Responed	jamith
10	2022-07-02	4 - atlas supplier	Not Responed	jamith

Product Id	Product Name	Item Code	Quantity
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Request for Quotation

Quate No: 8 Date : July 2, 2022

Comments:

Item Code	Description	Quantity
ka585	choclet	40

➤ Purchase order invoices

XPOS

JAMITH

2022-07-05 06:00:49

DASHBOARD

SUPPLIER

PRODUCT DETAILS

CUSTOMER

HUMAN RESOURCE

ADDITIONAL COST

EXTRA INCOME

SUPPLIER PRODUCT

INVOICES

REQUEST FOR QUATE

PURCHASE ORDER

RETURN

PURCHASE

SALE

INVOICES

SALE INVOICE

PURCHASE INVOICE

RETURN INVOICE

QUATE REQUEST

PURCHASE ORDER

PURCHASE ORDERS

Invoice Number

Search PO Number

Search Supplier

Search Supplier

PRINT INVOICE

ID	Supplier	Date	Sub Total	Discount	Final Total	Is Supplied
7	4 - atlas suppl.	2022-07-01	6500.0	0.0	6500.0	1
8	16 - kothmale	2022-07-02	11600.0	2320.0	9280.0	1
9	15 - kandos	2022-07-02	5500.0	0.0	5500.0	1
10	16 - kothmale	2022-07-02	3475.0	0.0	3475.0	1
11	4 - atlas suppl.	2022-07-02	3000.0	0.0	3000.0	1

ID	Product	Item Code	Item Price	Quantity	Total Price
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XPOS
11 Sea Street,
Colombo
0785423695
xpos@email.com

Purchase Order

Quate No: 10 Date : July 2, 2022

Supplier :

Name kothmale
Address colombo
Tele. No 0778523645
Reg. No 915357

Comments:

Quantity	Description	Unit Price	Total
20	choclet milk	60.00	1,200.00
35	vanila milk	65.00	2,275.00

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Request For Quate Interface

Before placing a purchase order, the required prices must be called and quotations obtained. The quotations are managed through this interface.

REQUEST FOR QUATE

Supplier:

Product ID: Brand: In Stock Quantity: Item Code: **ADD TO TABLE**

Description: Ordered Quantity: Quantity:

Quote Requesting Products

ID	Brand	Description	Item Code	Quantity

Purchase Actions: **DELETE ITEM**, **CANCEL ORDER**, **COMPLETE PURCHASE ORDER**

Purchase Required Products

Search:

ID	Brand	Description	Category	Quantity (In Stock)	Re-Order Level	Ordered Quantity
7	Atlas	single rule	School Item	48	100	0
8	Ice cream elephant house	chocolat ice cream	Ice cream	0	20	0
9	Ice cream elephant house	vanilla ice cream	Ice cream	0	20	0
10	coca cola	coca cola	soft drinks	0	10	0
11	coca cola	sprite	soft drinks	0	10	0
15	maliban bicat	maari	biscuits	0	60	0
16	madec rhikan	rhikan	meat	0	30	0

Purchase Order Interface

Purchase Order Interface is using the suppliers who gave the quotations, to provide quotes to the quotation to get enough products to the utility of less benefits, and put the purchase Order to the relevant Supplier.

PURCHASE ORDER

Supplier:

Product Id: Description: In Stock Quantity: Item Code: **ADD TO TABLE**

Brand: Purchase Price: Quantity: Total:

Purchased Products

ID	Brand	Description	Item Code	Purchase Price	Quantity	Total

Purchase Actions: **DELETE ITEM**, **CANCEL PURCHASE**, **COMPLETE PURCHASE**

Total Price: 0.00
Discount: 0
Discount Amount: 0.00
Total: **0.00**

Products

Description:

ID	Category	Brand	Description	Quantity (In Stock)	Re-Order Level	Ordered Quantity	Item Code

Return Interface

In this interface return Sales Handle after making a sale.

Purchase Interface

After leaving a purchase order, to give a good receiving notice to Supplier, stock update, Price Level manipulate are done in this interface.

Invoice generating on the product directly to the Customer, Balance is accurately generated, Stock updates are done in this interface.

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2. Software Interfaces

- The software will be used on PCs.
- The software interface will be developed by java programming language with NetBeans
- The software will run on JDK 8 environment.
- Personal database model will support MYSQL 8 environment as DBMS.
- Reports will generate using Japer soft studio

3. Hardware Interfaces

- Computer –
 - 1GB or higher RAM
 - 1Ghz or higher CPU
 - 40GB or higher HDD
 - Windows 7 or higher operating systems
- Keyboard
- Monitor
- Mouse
- Pos printer
- Printer
- Barcode reader



Other Non-functional Requirements

Safety Requirements

The user will have to enter the username and password so the system logged in using those details and system uses MD5 encryption for security. Then logged in system identifies access privileges for system for each user. And validate all form data.

Admin can access and control the all program in the system. But user cannot access the all program.

Usability Requirements

This system is easy to use by both admin and users such that they do not need to read an extensive amount of manuals.

This system is update at all the time in each of day.

Maintainability Requirements

The software will be composed of various modules decreasing the complexity of expansion.

Performance Requirements

The typing frequency should decrease to rise up the speed of the certain program.